

GHANA COMMUNICATION TECHNOLOGY UNIVERSITY



FACULTY OF COMPUTING AND INFORMATION SYSTEMS
DEPARTMENT OF INFORMATION TECHNOLOGY

BSc. (INFORMATION TECHNOLOGY & NETWORKING AND SYSTEMS
ADMINISTRATION) - LEVEL 100
EVENING / WEEKEND

END OF SEMESTER EXAMINATION, APRIL 2025

MATH 121: ALGEBRA

Duration: 2 Hours

INDEX NUMBER:

Instructions

1. Be sure to fill in your index number on both the question paper and answer booklet before starting the exam.
2. Answer *all questions* in SECTION A and *only two questions* in SECTION B.
3. Read all questions and possible answers *carefully*.
4. Both SECTION A and B must be answered in the answer booklet.

Submit Both the Question Paper and the Answer Booklet.

SECTION A: (MCQ'S) (ANSWER ALL QUESTIONS) [20 MARKS]

1. Solve for x : $\frac{1}{2}|2x + 8| - 10 = -2$
 - A. $x = 12$
 - B. $x = 8$ and $x = 4$
 - C. $x = 8$ and $x = -12$
 - D. There is no solution
2. Solve the inequality: $\frac{1}{3}|x + 4| - 5 < 2$
 - A. $(-25, 17)$
 - B. $(5, 10)$
 - C. $[-2, -17]$
 - D. $(-4, -2)$
3. Describe these two lines: $2x - y = 4$ and $6x - 3y = 9$?
 - A. Parallel
 - B. Perpendicular
 - C. Horizontal
 - D. None of these
4. What are the intercepts for $y = x^2 + 4x - 21$?
 - A. The x -intercept is -7 , there is no y -intercept.
 - B. The x -intercept is -3 , and the y -intercept 21 .
 - C. The x -intercepts are 7 and -3 , and the y -intercept is -21 .
 - D. The x -intercepts are 3 and -7 , and the y -intercept is -21 .
5. Rewrite $t = \log_b m$ as an exponential equation?
 - A. $t = m^b$
 - B. $b^m = t$
 - C. $m = b^t$
 - D. $m^t = b$
6. Expand the logarithm function: $\ln(\sqrt{x}y^2/\sqrt[3]{z})$?
 - A. $\frac{1}{2}\ln x + 2\ln y - \frac{1}{3}\ln z$
 - B. $\frac{\ln x^2 - 2\ln y}{\ln z}$
 - C. $\ln 2x + \ln 2y - \ln z$
 - D. $\ln x^2 + \ln y^2 - \ln z^3$

7. If $A = \begin{vmatrix} 3 & 4 \\ 2 & 3 \end{vmatrix}$, $B = \begin{vmatrix} -2 & -2 \\ 0 & -1 \end{vmatrix}$ then $(AB^T)^{-1}$ is:
- A. $\begin{vmatrix} -\frac{3}{2} & 2 \\ 5 & -7 \end{vmatrix}$
 B. does not exist.
 C. is a skew - symmetric.
 D. none of these.
8. If A is a 3×3 matrix for which $\det(A) = 7$ then $\det(4A) = ?$
- A. 448
 B. 343
 C. 1372
 D. 16384
9. Is the sequence an arithmetic progression? Find the number of terms: 3, 7, 11, ... if the last term is 123?
- A. yes, 31
 B. yes, -13
 C. yes, 13
 D. no
10. Find the sum of the infinite geometric series: $1 + \frac{1}{3} + \frac{1}{9} + \frac{1}{27} + \dots ?$
- A. $\frac{3}{2}$
 B. $\frac{2}{3}$
 C. $\frac{1}{2}$
 D. $\frac{1}{3}$

SECTION B (ANSWER ANY 2): [40 MARKS]

11. a) Sketch the solutions to the inequalities:
$$\begin{cases} x + y \leq 8 \\ x + y \geq 4 \\ x \leq 5; y \leq 5 \\ x \geq 0; y \geq 0 \end{cases}$$
 [16 marks]
- b) Ten years ago Juliet was four times as old as Becky. Now she is only twice as old as Becky. Find their present ages? [4 marks]
12. a) Solve for x: $5^x = 6^{x-1}$? [6 marks]
- b) Express as an algebraic sum of logarithms: $\log \frac{\sqrt{x^5}}{5\sqrt{y^2}}$? [6 marks]
- c) Evaluate the following logs: $\log 2$ and $\log 3$:
Express $\log 2.4$ in the above logs and evaluate it to four decimal places? [8 marks]
13. a) (i) Write the augmented matrix of the system of equation below: [4 marks]
- (ii) Use the Crammer's Method to solve the system of equation?
$$\begin{cases} x + y + z = -3 \\ 3x + y - 4z = 11 \\ 2y - 3y + 2z = 9 \end{cases}$$
 [6 marks]
- b) Use the Pascal triangle to expand the binomial expression:
 $(a - b)^7$ [10 marks]

ALGEBRAIC EXPRESSION		LINEAR EQUATIONS	
1	ADDITION:	1	$m = \frac{y_2 - y_1}{x_2 - x_1}$
	Add/Plus/Sum/Total/Increased by/More than	2	$Ax + By = C$
2	SUBTRACTION:	3	$y = mx + c$
	Minus/Difference/Subtract/Less than/Less	4	$y - y_1 = m(x - x_1)$
3	MULTIPLICATION: Product/Times/Multiply		
4	DIVISION: Quotient/Divide		
		QUADRATIC EQUATIONS	
		1	$ax^2 + bx + c = 0$
		2	$x^2 + (\text{sum of } aac)x + (\text{product of } ac) = 0$
1	$R(x) = \frac{P(x)}{Q(x)}$	3	$\text{Vertex} = \left(-\frac{b}{2a}, f\left(\frac{-b}{2a}\right)\right)$
2	V.A: $Q(x) = 0$; H.A: $\text{Deg}(P(x))/\text{Deg}(Q(x))$; Slant	4	$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$
3	Holes / Zeros		
		MATRICES	
	LOGARITHM	1	$A \pm B = \begin{bmatrix} a & b \\ c & d \end{bmatrix} \pm \begin{bmatrix} e & f \\ g & h \end{bmatrix} = \begin{bmatrix} a \pm e & b \pm f \\ c \pm g & d \pm h \end{bmatrix}$
1	$\log_a xy = \log_a x + \log_a y$	2	$AB = \begin{bmatrix} a & b \\ c & d \end{bmatrix} \begin{bmatrix} e & f \\ g & h \end{bmatrix} = \begin{bmatrix} ae + bg & af + bh \\ ce + dg & cf + dh \end{bmatrix}$
2	$\log_a \left(\frac{x}{y}\right) = \log_a x - \log_a y$	3	$kA = \begin{bmatrix} ka & kb \\ kc & kd \end{bmatrix}$
3	$\log_a x^n = n \log_a x$	4	$\det(kA) = k^n \cdot \det(A)$
4	$\log_a x = \frac{\log_c x}{\log_c a}$	5	$\det(AB) = \det(A) \cdot \det(B)$
5	$\log_b 1 = 0 \qquad \log_b b = 1$	6	$\det(A)^T = \det(A)$
6	$\log_{b^n} a^m = \frac{m}{n} \log_b a$	7	$\det(A^{-1}) = \frac{1}{\det(A)}$
7	$\ln = \log_e x$	8	$AX = B \qquad X = A^{-1}B$
8	$\log_a x = y \Rightarrow x = a^y$	9	$A^{-1} = \frac{1}{\det(A)} \text{Adj}(A)$
SEQUENCES AND SERIES		COMBINATORICS	
1	$l = a + (n - 1) d$	1	AND = X; OR = +
2	$S = \frac{n}{2} (a + l) = \frac{n}{2} [2a + (n - 1)d]$	2	Straight line: N!

3	$l = ar^{n-1}$	3	Circular: $(N-1)!$
4	$S = \frac{a(r^n - 1)}{r - 1}$	4	Permutation: $P(n, k) = \frac{n!}{(n-k)!}$
5	$s_{\infty} = \frac{a}{1-r}, r < 1$	5	Combination: $C(n, k) = \frac{n!}{(n-k)!k!}$
	INDUCTION	6	Rearrangement: $\frac{n!}{n_1! n_2! \dots n_k!}$
1	Show for $n = 1$ and $n = 0$: $P(1)$ & $P(0)$ is true		
2	Assume $n = k$, $P(k)$ is true	7	Coefficient of $x^3y^6 = C(9, 6)$
3	Show for $n = k + 1$, $P(k + 1)$ is also true		
4	Conclude it is True for all n		